Harshwardhan Patil

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Summary

Full-Stack Engineer with **3 years of experience** building scalable, distributed systems and high-performance APIs across frontend, backend, and cloud. Skilled in **system design**, CI/CD on AWS, production monitoring, and delivering **fault-tolerant, data-consistent services** at scale in **Agile** environments.

Technical Skills

- **Languages & Frameworks:** Java, TypeScript, JavaScript, SQL, Spring Boot, Angular, Node.js, Express.js, Next.js, WebSocket, Python, FastAPI
- **Cloud & DevOps:** AWS (EC2, ECS, S3, RDS, CloudFront, ALB, SSM, Lambda), Docker, Jenkins, Git, GitHub Actions, Nginx, Cloudflare, Kafka
- **Databases:** PostgreSQL, Redis, MySQL, Neo4j, Elasticsearch, TimescaleDB, Schema design, Database indexing, Query optimization, Caching strategies
- **Testing & Security:** JUnit, Mockito, Jasmine, Vitest, ESLint, SonarQube, Postman, JWT, RBAC, HTTPS/TLS, CORS, CSRF/XSS protection
- **Design & Architecture:** System Design, Design Patterns, SOLID principles, Distributed backend services, Concurrency handling, Session management
- **AI & ML:** Spring AI, OpenAI API, Prompt Engineering, Embeddings, Vector Search, RAG Pipelines, pgvector, ChromaDB, Tool Calling, Pydantic

Professional Experience

Software Engineer, Clarivate Analytics – Bangalore, India *Aug 2023 - Jul 2024*

- Developed scalable **Spring Boot** REST APIs with DTO contracts and standardized error handling, reducing breaking changes by **30%** across releases.
- Implemented **JWT**-based multi-session management across backend services, capping concurrent logins at 12 per user with token invalidation on breach.
- Optimized **PostgreSQL** schemas with **composite indexing** across reporting and search workloads for drug and patent datasets, cutting full table scans.
- Containerized and deployed microservices to **EC2** using **Docker** and **Jenkins CI/CD** pipelines, supporting **50+ production releases** with zero rollbacks.
- Enforced **role-based access control** across backend and frontend layers, applying tiered and region-based data visibility for **10+ access variants**.
- Implemented low-latency **Elasticsearch**-powered typeahead and autocomplete, improving search response times by **40%** over large-scale datasets.
- Developed dynamic PPT and PDF export pipelines using **PptxGenJS** and **jsPDF**, generating **100+ customized reports** with user-selected filters.
- Optimized **PostgreSQL** query logic for PPT export, decoupling bulk multi-filter chart fetching from single-chart UI rendering to reduce DB round-trips.
- Achieved **90%** test coverage using **JUnit** and **Mockito** for backend and **Jasmine** for Angular, covering unit, branch, and line coverage across all modules.

Associate Software Engineer, Clarivate Analytics – Bangalore, India *Aug 2021 - Jul 2023*

- Integrated **Elasticsearch** into **Spring Boot**, enabling advanced full-text queries across **millions of drug records** and reducing search latency by **30%**.
- Migrated **5+ core business flows** from Angular 8 to Angular 13, rebuilding components with a modular architecture and improved state management.
- Built secure request-handling across backend services using **Spring Security**, **CORS**/**CSRF**, and input validation to guard against unauthorized access.
- Boosted Angular performance by **20–25%** via lazy loading, **RxJS** streams, **OnPush** change detection, **memoization**, and optimized data-fetching patterns.
- Built **D3.js** chart components with grouping and filtering aggregations in Angular, powering **15+ analytics views** across drug intelligence dashboards.
- Architected a configurable **Angular** table framework with sorting, pagination, and filters; adopted across **3 internal teams**, cutting duplication by **50%**.

Projects

TinyURL — URL Shortener *Feb 2026 - Mar 2026*

Production-grade URL shortening service with QR code generation, link expiry, and a fully automated cloud deployment pipeline.

Tools: Spring Boot, Angular, PostgreSQL, AWS (EC2, RDS, S3, CloudFront, ALB), Docker, Cloudflare, Nginx, Lambda, EventBridge, SQS DLQ, CloudWatch

- Designed 6-layer request path across Cloudflare, CloudFront, ALB, Nginx, **Spring Boot**, and **PostgreSQL**, separating SPA delivery from API and redirects.
- Hardened API traffic with **Nginx rate limits**, **Bucket4j token buckets**, **layer-7** bot/scanner filters, per-IP request size limits, and **HSTS**/**CSP** headers.
- Designed serverless EC2/RDS scheduler with **Lambda**, **EventBridge**, CloudWatch alarms, SNS, and **SQS DLQ**, reducing monthly cloud cost by **~36%**.
- Built zero-SSH **CI/CD** with GitHub Actions, AWS OIDC, SSM RunCommand, Docker smoke tests, cosign signing, and auto CloudFront cache invalidation.

diagram-sync — Diagram-as-Code CLI 4,816 *May 2026 - Present*

Published npm package that renders architecture diagrams from 7 source formats into versioned SVG/PNG outputs via `npx diagram-sync`.

Tools: TypeScript, Node.js, Vitest, ESLint, GitHub Actions

- Engineered **provider-registry architecture** mapping 11 file extensions to isolated rendering adapters with runtime graceful degradation on missing tools.
- Developed recursive repo scan, `--files`, and `--changed` via `git diff HEAD` with untracked support, enabling CI to regenerate only PR-changed files.
- Architected **tiered format resolution** across CLI flag, job, and global config with per-provider validation and actionable errors for unsupported formats.

Sentinel — Real-Time Geospatial Anomaly Detection *Aug 2026 - Present*

Distributed platform ingesting ADS-B/AIS telemetry to track entity state, detect rule-based anomalies, and surface correlated alerts on a live map.

Tools: TypeScript, Node.js, Kafka/Redpanda, TimescaleDB, Neo4j, Redis, Express, WebSocket, Next.js

- Decoupled telemetry ingestion from anomaly evaluation via 4 **Kafka**/**Redpanda** pipelines, letting deviation and proximity detection scale independently.
- Designed **replay-safe idempotency** without exactly-once transport, using deterministic alert/position IDs and **transactional upsert** into **TimescaleDB**.
- Enabled real-time proximity alerts across thousands of entities by pre-filtering candidates with **Uber H3** geospatial indexing before exact distance checks.
- Coordinated a single active alert evaluator via a **Redis leader lease**, preventing concurrent evaluation while preserving replay safety with deterministic IDs.

smart-anytool-agent — Research-Backed Tool-Calling Agent *Jun 2026 - Present*

Implements SMART and AnyTool techniques with self-awareness gating, hierarchical tool filtering, and self-reflection retry loops.

Tools: Python, OpenAI API, Pydantic, httpx, GitHub Actions

- Built **SMART self-awareness** LLM pre-check gate; in-house A/B eval showed **98%** routing accuracy, **49%** lower latency and tokens on knowledge queries.
- Engineered **hierarchical tool filtering** with LLM category classification and progressive widening fallback, exposing only relevant tool subsets per query.
- Built **self-reflection recovery loop** with parallel async tool execution and **Pydantic**-validated dispatch, handling multi-step agentic queries end-to-end.

Education

University of Texas at Arlington *Master's in Software Engineering* *GPA: 3.6 / 4.0* *Aug 2024 - May 2026*